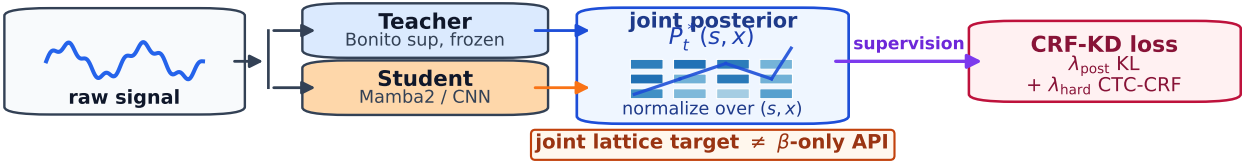
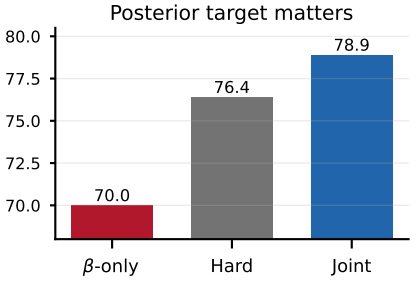
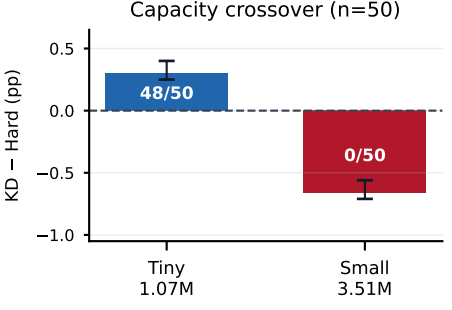
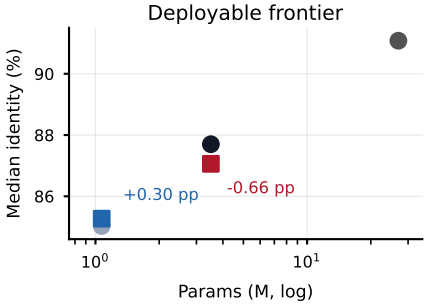

raw signal
Teacher
Bonito sup, frozen
Student
Mamba2 / CNN
joint posterior
$P_t(s, x)$
normalize over (s, x)
supervision
CRF-KD loss
λ_{post} KL
+ λ_{hard} CTC-CRF
joint lattice target ≠ β-only API


Posterior target matters
80.0
77.5
75.0
72.5
70.0
70.0
76.4
78.9
β-only
Hard
Joint


Capacity crossover (n=50)
KD − Hard (pp)
0.5
0.0
−0.5
−1.0
48/50
0/50
Tiny
1.07M
Small
3.51M


Deployable frontier
Median identity (%)
90
88
86
+0.30 pp
-0.66 pp
10^0
10^1
Params (M, log)